



PROJECT VBT-X1

A National First in Wearable Performance & Rehabilitation Technology

"We are building, end-to-end and inside our own faculty, a wearable medical-grade sensor that will redefine how athletes train, how patients rehabilitate, and how the human body is measured in motion."

Project VBT-X1, Faculty of Electronic Engineering

AT A GLANCE

A coin-sized wearable sports sensor — designed, built, and validated *entirely inside our faculty* — that brings hospital-grade movement analysis to athletes, surgeons, and physiotherapists.

Built for Sports medicine, clinical rehabilitation, public health

Status Active data acquisition with volunteer subjects

Ambition The first university-fabricated sports sensor in our nation



The VBT-X1 concept in real life.

THE VISION

Project VBT-X1 is a faculty-led initiative to design, build, and clinically validate a postage-stamp-sized intelligent sensor that measures human movement with the precision of a hospital laboratory — and the convenience of a smart watch. Worn discreetly during training, physical therapy, or post-surgical rehabilitation, it provides physicians and coaches with real-time, objective data on muscular performance, fatigue, and recovery.

Every component — from the sensor electronics to the custom silicon chip at its heart — is being designed and built by our students and faculty.

IMPACT Where VBT-X1 Changes Lives

- ◆ **Sports Medicine.** Objective fatigue and overtraining detection for national-level athletes and Olympic candidates.
- + **Clinical Rehabilitation.** Quantitative recovery benchmarks after ACL, rotator cuff, and orthopaedic surgery.
- **Public Health.** Bringing laboratory-grade movement assessment to gyms, clinics, and physiotherapy rooms nationwide.
- **National Technology Sovereignty.** A locally designed semiconductor product — a first for our university.

WHY WE LEAD

Few institutions worldwide attempt this full vertical: sensors, embedded software, machine intelligence, and custom microchip design — all under one academic roof. Our team has assembled the rarely-combined expertise to take a clinical idea from the laboratory bench to a fabricated chip, with international research partners observing our progress.

BY THE NUMBERS

50+

VOLUNTEER SUBJECTS

From across the university community

12,480

MEASURED REPETITIONS

Captured under laboratory conditions

99.2%

TRACKING ACCURACY

Validated against optical reference

< 1 mW

TARGET POWER

Smaller than a hearing-aid battery

2

INTERNATIONAL PARTNERS

Research collaborations underway

1

PATENTS

Filings in preparation

MILESTONES

- Q2 2026 Data acquisition campaign (*in progress*)
- Q2 2026 Intelligence model validated in clinic
- Q3 2026 Hardware prototype demonstration

IN PARTNERSHIP WITH



A Faculty Investment in National Leadership.

From our laboratories to the world's clinics, gyms, and Olympic training centres.

YOUTH ADMINISTRATION OFFICE

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